Implementation Report

1. Introduction

The objective of this design is to implement a specialized arithmetic unit that performs the addition of two numbers represented in **Q2.5 fixed-point format** and subsequently calculates the **absolute value (magnitude)** of the resulting sum. This module is essential for applications such as digital signal processing (DSP), error analysis, and magnitude detection where the sign of the result is less important than its distance from zero.

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2. Numerical Representation Specification

2.1 The Q2.5 Format

The design utilizes an 8-bit fixed-point representation known as Q2.5. This format is a subset of the Two's Complement system, partitioned as follows:

- **Total Word Length:** 8 bits.
- **Sign Bit (S):** 1 bit (Most Significant Bit - MSB).
- **Integer Portion (I):** 2 bits.
- **Fractional Portion (F):** 5 bits.

Bit Index	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Weight	-2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}

2.2 Mathematical Characteristics

- **Resolution (LSB):** $2^{-5} = 0.03125$. This is the smallest increment the system can represent.
- **Dynamic Range:**
    - **Maximum Positive:** $011.11111_2 = (2^2 - 2^{-5}) = \mathbf{3.96875}$.
    - **Minimum Negative:** $100.00000_2 = -2^2 = \mathbf{-4.00000}$.

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3. Hardware Architecture and Logic Design

The module `q2_5_adder_abs` is implemented using combinational logic to ensure high-speed processing. The architecture follows a three-stage internal pipeline:

3.1 Stage 1: Extended Signed Addition

To prevent silent overflow (where a carry bit is lost), the module performs addition using a **9-bit intermediate register**(`temp_sum`).

- **Sign Extension:** The 8-bit inputs `a_in` and `b_in` are sign-extended to 9 bits before addition. This ensures that if the sum of two positive numbers exceeds 3.96875, the carry bit is captured in the 9th bit rather than being truncated.

3.2 Stage 2: Overflow Detection and Saturation

Because the output must fit back into an 8-bit format, the module checks if the `temp_sum` resides outside the valid Q2.5 range ($[-128, 127]$ in raw integer steps).

- If the sum exceeds the range, an `overflow` flag is asserted.
- **Saturation Logic:** In the event of an overflow, the output is "clamped" or saturated to the maximum representable value (`8'h7F`), preventing the mathematically incorrect "wrap-around" effect typical of standard binary adders.

3.3 Stage 3: Absolute Value Computation

The absolute value logic implements the mathematical function $|x|$:

1. **If Sum ≥ 0:** The magnitude is identical to the sum. The output takes the lower 8 bits of the `temp_sum`.
2. **If Sum < 0:** The module performs a **Two's Complement Negation**. This involves inverting all bits of the sum and adding 1 ($|x| = \sim x + 1$). This converts the negative value to its positive equivalent.

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4. Input/Output Port Description

Port Name	Direction	Width	Description
`a_in`	Input	8 bits	Signed Q2.5 operand A.
`b_in`	Input	8 bits	Signed Q2.5 operand B.

`abs_sum_out`	Output	8 bits	The absolute magnitude $
`overflow`	Output	1 bit	Active-high signal indicating result exceeded Q2.5 range.

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5. Error and Precision Analysis

5.1 Quantization Error

Since the inputs and the internal arithmetic are all performed in the same Q2.5 scale, no additional quantization error is introduced during the addition or absolute value stages. The error relative to an ideal infinite-precision real number is capped at $\pm LSB/2$ (≈ 0.0156).

5.2 Asymmetric Absolute Value Case

A unique edge case in Two's Complement is the value -4.00 (`8'h80`). The absolute value of -4.00 is $+4.00$. However, in Q2.5, the maximum positive value is 3.96875.

- **Design Choice:** Our implementation handles this by saturating the absolute value of -4.00 to 3.96875 and asserting the `overflow` flag, ensuring the system remains stable and does not produce a negative magnitude.

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6. Conclusion

The `q2_5_adder_abs` module provides a robust solution for fixed-point magnitude calculations. By incorporating sign extension, overflow detection, and saturation logic, it maintains mathematical integrity even when results exceed the narrow dynamic range of the 8-bit Q2.5 format. This design is highly resource-efficient, utilizing only standard adders and multiplexer-based negation logic, making it ideal for FPGA-based DSP architectures.